Univariate Linear Regression

Fundamentals | Notation, Model, and the Path to a Cost Function

1. Introduction to Linear Regression

Linear regression is arguably the most widely used learning algorithm in the world today. It's the common starting point because it establishes the core principles of predictive modeling: at its simplest, the process is "fitting a straight line" to a dataset to find the underlying pattern.

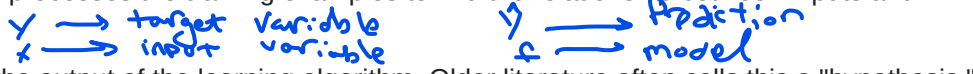
Worked Example - Portland Real Estate

As a real estate agent, you plot house size (sq ft, x-axis) against selling price in thousands of dollars (y-axis) using historical data. To estimate the value of a 1,250 sq ft house not in your dataset, you find where that size intersects the "best-fit" line - yielding a predicted price of roughly $220,000.

2. The Supervised Learning Workflow

This is called "supervised" learning because the algorithm trains on a dataset that already contains the "right answers" - the actual historical prices for specific house sizes. The workflow follows a logical progression:

- **Training Set** - the input data: features (e.g. house size) and targets (e.g. actual selling price y).
- **Learning Algorithm** - processes the training examples to find the relationship between inputs and outputs.
- **Function/Model (f)** - the output of the learning algorithm. Older literature often calls this a "hypothesis."
- **Prediction ($\hat{y}$)** - given a new input x, the model outputs an estimate, pronounced "y-hat."



Key distinction:

y is the actual "right answer" from the training data. $\hat{y}$ (y-hat) is the model's estimate, which may or may not match the true value exactly.

3. Categorization: Regression vs. Classification

Choosing the right category depends entirely on the nature of the output you want to predict.

Model Type	Output Nature	Example

Regression	Infinitely many possible continuous numerical values	Predicting house price ($220,000, $221,500.50)
Classification	Discrete categories from a small, finite set	"Cat" vs "dog"; diagnosing a specific disease

Regression addresses problems with numerical outputs across a spectrum; classification deals with a limited number of possible outcomes.

4. Standard Machine Learning Notation

A shared mathematical language keeps communication precise across the field. Reference this table often - illustrated using the first row of the Portland dataset, where a 2,104 sq ft house sold for $400,000.

Symbol	Name	Description	Portland Example
x	Input Variable	Feature used to make a prediction (e.g. size)	x = 2104
y	Output Variable	Target variable / "right answer" (thousands of $)	y = 400
m	Number of Examples	Total rows/data points in the training set	m = 47
(x, y)	Training Example	A single input-output pair	(2104, 400)
$(x^{(i)}, y^{(i)})$	i-th Training Example	Index referring to a specific row (1 to m)	$(x^{(1)}, y^{(1)})$
$\hat{y}$	Prediction	The estimate produced by model f	$\hat{y} \approx 220$

Important note on indexing:

The superscript (i) in $(x^{(i)}, y^{(i)})$ is an **index** identifying a specific row in the series $x^{(1)}, x^{(2)}, ..., x^{(m)}$. It is NOT a mathematical exponent - don't square anything.

5. The Mathematical Representation

The univariate linear regression model is written as:

$$f_{w,b}(x) = wx + b$$

Often shortened to f(x). Breaking down the components:

- **Univariate** - from Latin *uni* (one) + *variate* (variable): the model uses a single input feature x.
- **w and b** - the model's parameters. w is the slope of the line; b is the y-intercept.

- **The Line** - w and b together set the exact position and angle of the line. Changing them changes the prediction $\hat{y}$ for any given x.

6. Conceptual Foundations: Why Linear Models?

Real data is often non-linear, and later models will fit complex curves. Linear regression is the starting point because it's highly efficient and easy to interpret.

> **Why this matters long-term:**
>
> The weights (w) and biases (b) introduced here are the exact same fundamental components used to build deep learning architectures and neural networks. Mastering the linear case is a prerequisite for understanding far more complex AI models.

7. Conclusion and Next Steps

Defining f_w,b(x) = wx + b is only the first step. The real challenge is finding the optimal values of w and b that produce the "best-fit" line for your specific training data.

To solve this, the next concept is the **Cost Function** - a universal idea in AI used to measure how far the model's predictions are from the actual "right answers." Minimizing this cost is how the algorithm "learns" the best parameters. Upcoming Python labs will have you define these functions in code and visualize how different w and b values affect accuracy.

Study guide synthesised from lecture transcripts - Univariate Linear Regression Fundamentals.